SMARTBRIDGE
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SKILL WALLET
A SmartBridge Product

Project Initialization and Planning Phase

Date	30 June 2026
Team ID	xxxxx
Project Title	Flood Prediction System
Maximum Marks	5 Marks

Project Proposal (Proposed Solution) report

The proposed solution aims to improve flood prediction using machine learning, enabling faster and more accurate identification of potential flood risks. The system addresses the limitations of traditional flood monitoring methods by providing intelligent, data-driven predictions based on historical weather data. This solution helps disaster management authorities and local communities prepare for flood events more effectively, reducing potential damage and improving public safety. Key features include an XGBoost-based flood prediction model, real-time prediction through a Flask web application, and an easy-to-use interface for entering weather parameters.

Project Overview	
Objective	The primary objective is to develop an intelligent Flood Prediction System using Machine Learning that accurately predicts flood occurrences based on weather parameters, enabling early warning and effective disaster preparedness.
Scope	The project focuses on collecting and analyzing historical weather data, training multiple machine learning models, selecting the best-performing model, and deploying it through a Flask web application for real-time flood prediction.
Problem Statement	
Description	Traditional flood prediction methods often rely on manual monitoring and delayed analysis, making it difficult to provide timely and accurate flood warnings. These limitations increase the risk of property damage, environmental loss, and threats to human life.
Impact	Developing an accurate flood prediction system enables early warning, improves disaster preparedness, supports better resource allocation, reduces economic losses, and enhances public safety during flood events.
Proposed Solution	
Approach	Develop a machine learning-based flood prediction system that analyzes weather-related parameters such as cloud cover and rainfall to classify flood risk. The trained model is integrated with a Flask web application to provide instant predictions based on user input.

Key Features	• Machine learning-based flood prediction using XGBoost. • Real-time prediction through a Flask web application. • User-friendly interface for entering weather parameters. • Data preprocessing and feature scaling for improved prediction accuracy. • Fast and reliable prediction results to support disaster management.

Resource Requirements:

Resource Type	**Description**	**Specification/Allocation**
Hardware		
Computing Resources	CPU/GPU specifications, number of cores	Intel Core i5/i7 Processor (or equivalent)
Memory	RAM specifications	8 GB
Storage	Disk space for data, models, and logs	256 GB SSD or higher
Software		
Frameworks	Python frameworks	Flask
Libraries	Additional libraries	scikit-learn, XGBoost, pandas, numpy, matplotlib, seaborn, joblib
Development Environment	IDE	Jupyter Notebook, Visual Studio Code
Data		
Data	Source, size, format	Kaggle Flood Prediction Dataset, 115 records, Excel/CSV format containing weather-related features such as Cloud Cover, Annual Rainfall, Jan–Feb Rainfall, Mar–May Rainfall, Jun–Sep Rainfall, and the target Flood label.